



QUANTITATIVE METHODS 1 · CHAPTER 2

Measures of Central Tendency

Exercise series

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1. Mean, median and mode

Waiting times are 4, 5, 5, 6, 7, 8, 9, 26 minutes. Calculate the mode, median and mean. Which measure best describes a typical wait?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

2. Weighted decisions

A course grade combines quiz 20%, project 30% and final examination 50%. A student scores 14, 16 and 11 out of 20.

- 1 Calculate the weighted mean.
- 2 Compare it with the unweighted mean.
- 3 Explain why the two differ.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

3. Grouped median and quartiles

Order values are grouped: $[0,20[: 8$; $[20,40[: 18$; $[40,60[: 24$; $[60,100[: 10$.
Estimate Q_1 , the median and Q_3 by linear interpolation.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

4. Geometric and harmonic means

- 1 Revenue changes by +20%, -10% and +15%. Find the equivalent average annual growth rate.
- 2 A driver covers two equal distances at 60 and 90 km/h. Find the average speed.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

5. Choosing the centre

For each case choose a principal measure and justify it: employee salaries with a few executives; most common product size; average portfolio multiplier over four years; average price paid when quantities differ; satisfaction categories from very dissatisfied to very satisfied.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

6. Robust centre for delivery performance

Delivery times (minutes) for 12 orders are: 18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 65.

- 1 Calculate the mode, median and mean.
- 2 Recalculate the mean without the 65-minute delivery.
- 3 Recommend the principal indicator for a customer-facing dashboard and justify your choice.

Method prompt. Compare the indicators before and after removing the unusual observation; do not delete it without explanation.

7. Integrated centre-selection case

A retailer reports three indicators:

Question	Data
Average basket value	120 baskets: €3,600 total
Typical annual growth	+12%, -8%, +15%, +5%
Average speed	Two equal 90-km legs at 60 and 90 km/h

- 1 Choose and calculate the appropriate mean for each row.
- 2 Explain why using the arithmetic mean in all three cases is incorrect.
- 3 Write a short managerial conclusion.

Method prompt. Match additive totals, compounded factors and equal-distance rates with arithmetic, geometric and harmonic means.